

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 30%

Dr DUMPALA PRASANTH

QUESTIONS: 10

Total Marks: 50

Annexure A

CODING CHALLENGE

Duration: 2hrs

1. Predict the Reinforcement Learning (RL) framework by describing the agent–environment interaction, emphasizing the importance of states, actions, and rewards, and discuss how cumulative reward influences decision-making. **(5 marks)**

```
import random
state = 0
for step in range(10):
    action = random.choice([-1, 1])
    next_state = state + action
    reward = 1 if next_state == 5 else -1
    print("State:", state, "Action:", action,
          "Reward:", reward, "Next:", next_state)
    state = next_state
```

2. Estimate how decision-making problems can be modeled as a Markov Decision Process (MDP), detailing the components of action space, transition probability, reward function, and discount factor. **(5 marks)**

```

import random
states = [0, 1, 2, 3, 4]
actions = [-1, 1]
gamma = 0.9
state = 0
for i in range(10):
    action = random.choice(actions)
    next_state = max(0, min(4, state + action))
    reward = 10 if next_state == 4 else -1
    print(state, action, reward, next_state)
    state = next_state

```

3. Discuss the role of policy and value functions in RL, and examine how state-value and action-value functions contribute to evaluating long-term benefits of actions using the Bellman Equation. **(5 marks)**

```

gamma = 0.9
reward = 10
next_value = 5

value = reward + gamma * next_value

print("Bellman Value:", value)

```

4. Justify the exploration–exploitation trade-off in RL and compare strategies such as ϵ -greedy and softmax approaches in balancing learning efficiency. **(5 marks)**

```

import random
import numpy as np

Q = [2, 5, 3]
epsilon = 0.1

if random.random() < epsilon:
    action = random.randrange(3)
else:
    action = np.argmax(Q)

print("Epsilon-Greedy Action:", action)

temperature = 1
prob = np.exp(Q / temperature)
prob /= sum(prob)

action = np.random.choice(3, p=prob)
print("Softmax Action:", action)

```

5. Explain how reward shaping can accelerate learning in RL and illustrate its impact on preventing suboptimal policies. **(5 marks)**

```

position = 0
goal = 5

for step in range(10):
    position += 1

    if position == goal:
        reward = 100
    else:
        reward = 5

    print("Position:", position, "Reward:", reward)

    if position == goal:
        break

```

6. Identify the challenges of applying RL in real-world robotics and analyze how safety, sample efficiency, and environment variability affect agent performance. **(5 marks)**

```

import random

position = 0
goal = 5

for step in range(10):
    action = random.choice([-1, 1])
    position += action

    if position < 0:
        position = 0

    reward = 100 if position == goal else -1

    print("Position:", position, "Action:", action, "Reward:", reward)

    if position == goal:
        print("Goal reached!")
        break

```

7. Analyze the influence of the discount factor (γ) on short-term versus long-term decision-making, and evaluate its role in episodic tasks. **(5 marks)**

```

reward = [10, 20, 30]
for gamma in [0.1, 0.5, 0.9, 0.99]:
    total = sum(r * gamma**i for i, r in enumerate(reward))
    print("Gamma:", gamma, "Return:", total)

```

8. Evaluate the effectiveness of model-free RL compared to model-based RL in dynamic environments, and differentiate their strengths and limitations. **(5 marks)**

```

import random

```

```
Q = {}
```

```
state = 0  
action = random.choice([0, 1])  
reward = 10
```

```
Q[(state, action)] = reward
```

```
print("Model-Free Q:", Q)
```

```
# Model-Based: store transition  
model = {}  
next_state = 1
```

```
model[(state, action)] = (next_state, reward)
```

```
print("Model-Based Model:", model)
```

9. Illustrate with an example how Q-learning updates the action-value function during training, and demonstrate its convergence behavior. **(5 marks)**

```
import random
```

```
Q = [[0, 0] for _ in range(5)]
```

```
alpha = 0.5  
gamma = 0.9
```

```
state = 0  
action = random.randint(0, 1)  
next_state = 1  
reward = 10
```

```
old = Q[state][action]
```

```
Q[state][action] = old + alpha * (  
    reward + gamma * max(Q[next_state]) - old  
)
```

```
print("Old Q-value:", old)  
print("New Q-value:", Q[state][action])
```

10 Differentiate between on-policy and off-policy learning methods in RL, and assess their advantages and limitations in practical applications. **(5 marks)**

```
alpha = 0.5  
gamma = 0.9
```

```
Q = [[0, 0] for _ in range(3)]
```

```

state = 0
action = 1
reward = 10
next_state = 1
next_action = 0

# SARSA - On Policy
Q[state][action] += alpha * (
    reward + gamma * Q[next_state][next_action]
    - Q[state][action]
)

print("SARSA:", Q)

# Q-Learning - Off Policy
Q[state][action] += alpha * (
    reward + gamma * max(Q[next_state])
    - Q[state][action]
)

print("Q-Learning:", Q)

```

Code of Conduct:

*I G Saswanth(192425150) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: G Saswanth

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
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Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately